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National Institute of Technology, Raipur (Chhattisgarh), India

Department of Information Technology

Minor Project Final Presentation — November 2026

Spatial-Guided Spectral-Channel Fusion Network for Hyperspectral Remote Sensing Image Analysis

Under the Guidance of

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National Institute of Technology, Raipur

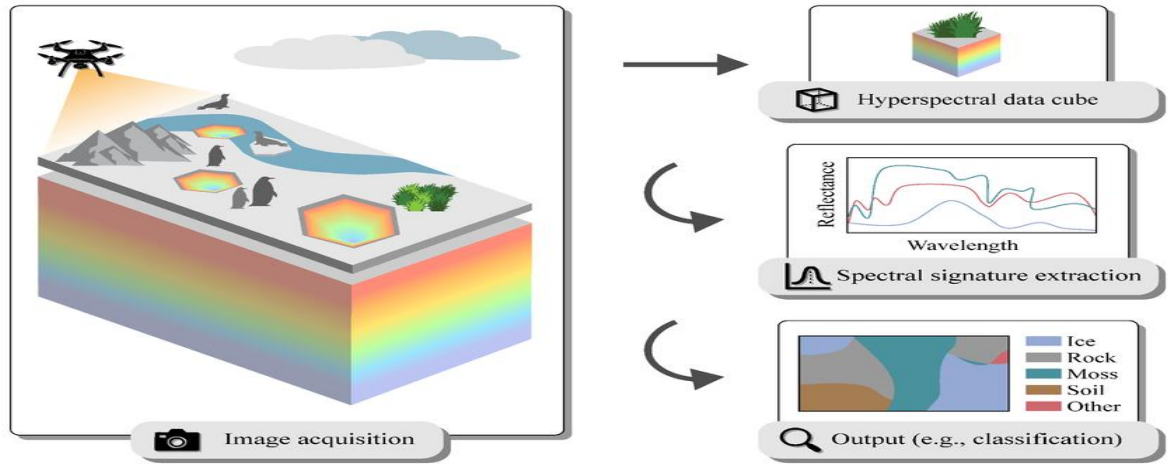
Outline

- Recap: Problem Statement & Research Objectives
- SGSCF-Net Architecture Overview
- Module Design: SBAM, CBSA, SCFG
- Hybrid Loss Function
- Experimental Setup & Benchmark Datasets
- Results: Indian Pines, Pavia University, Salinas Valley
- Classification Maps Comparison
- Computational Efficiency & Parameter Analysis
- Ablation Study & Depth Sensitivity
- Hyperparameter Grid Search (60 Runs)
- Data Scarcity Stress Testing (1%–10%)
- Cross-Dataset Summary & Key Achievements
- Future Work & Research Directions

Recap: Problem Statement & Research Objectives

Problems Identified (Mid-Semester)

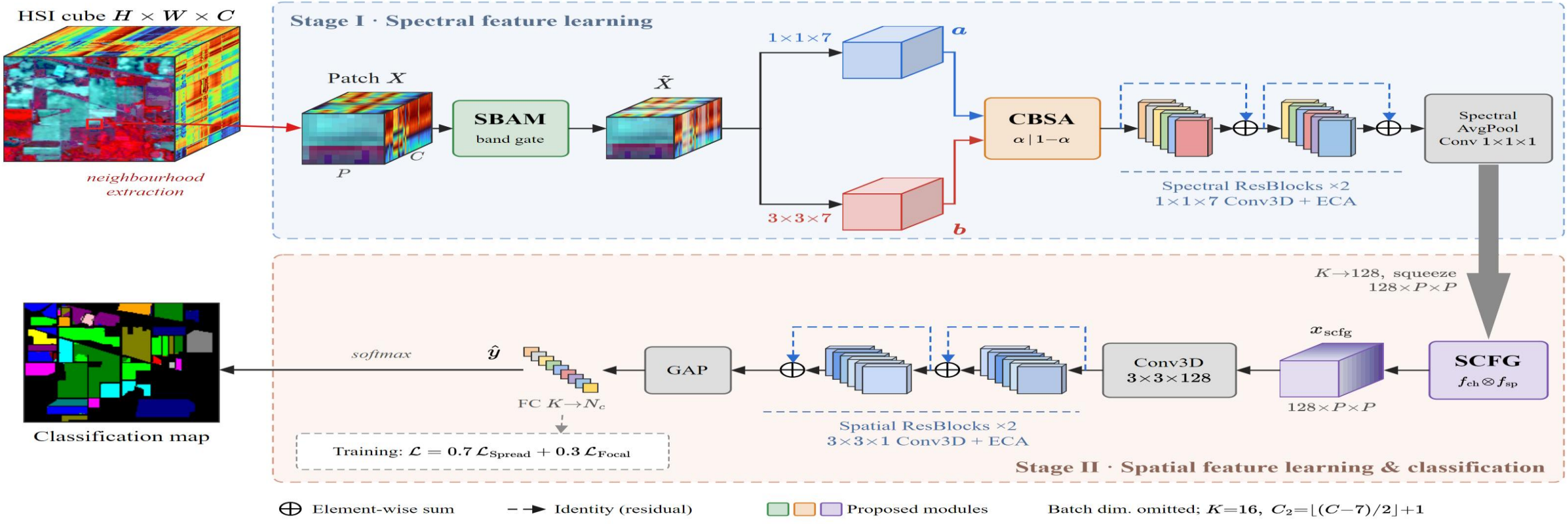
- Hughes Phenomenon: High bands + low samples = overfitting
- Memory Explosion: 3D-CNNs & Transformers exceed 85M+ parameters
- Feature Entanglement: Spatial + spectral mixing blurs class signatures
- Class Imbalance: Minority classes (Oats: 20 samples) collapse to 0%
- Unfiltered Noise: Atmospheric absorption corrupts raw band features



What We Delivered (This Presentation)

- Fully implemented & trained SGSCF-Net across 3 benchmark HSI datasets (Indian Pines, Pavia University, Salinas Valley)
- 120+ GPU training runs: 60-run patch/batch grid search + 12 data scarcity stress tests + 24 baseline benchmark runs
- Competitive or superior accuracy vs. 8 SOTA baselines while using only 308K parameters (vs. 6.4M for SSAN, ~610K for ViT)
- Complete 31-page ACM-format research paper with per-class breakdowns, classification maps, ablation studies, convergence analysis

SGSCF-Net: End-to-End Architecture Overview



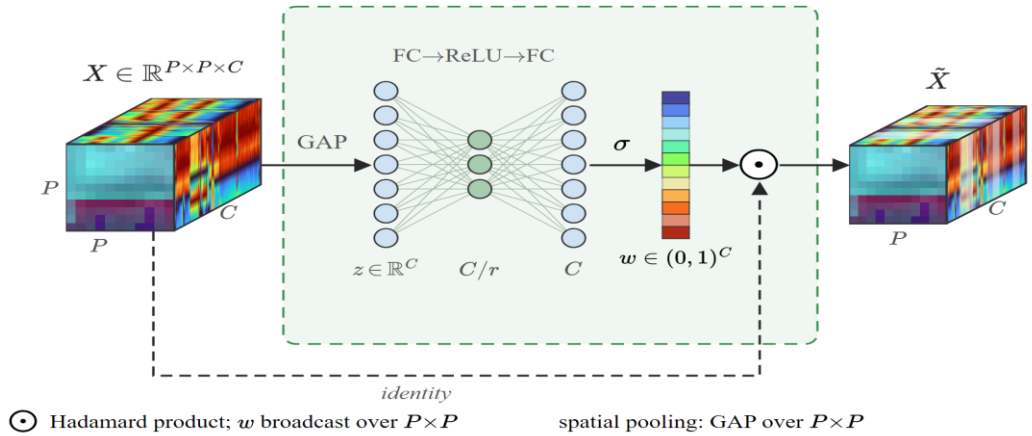
Input: X ∈ ℝ^(B×1×P×P×C) → SBAM recalibration → Dual-branch CBSA → Spectral deepening → SCFG covariance gating → Spatial refinement → ŷ ∈ ℝ^(B×C_cls)

Stage I: SBAM band gate → Stage II: Dual 3D entry + CBSA complementary gating → Spectral ResBlocks ×2 + ECA → Stage III: SCFG covariance transition → Stage IV: Spatial ResBlocks → GAP + FC | Loss: 0.7×Spread + 0.3×Focal

Module 1: Spectral Band Attention Module (SBAM)

What SBAM Does

- Operates on RAW spectral bands BEFORE any convolution
- Adaptive 3D average pooling compresses P×P spatial grid
- Squeeze MLP bottleneck (reduction r=8) learns band weights
- Sigmoid gate w ∈ (0,1)^C suppresses noisy atmospheric bands
- Hadamard product: $\tilde{X} = X \odot w$ (channel-wise reweighting)



Key Equations & Design Insight

- $z_c = (1/P^2) \sum X(b,1,u,v,c)$ [spatial pooling]
- $w = \sigma(W_2 \cdot \delta(W_1 z + b_1) + b_2)$ [MLP bottleneck]
- $\tilde{X} = X \odot E(w)$ [band filtering]
- Recalibrates BEFORE convolution (unlike SENet/CBAM)
- Only 10,225 params (3.32% of total network)

Diagram: SBAM Flow

- Raw input X passes through global average pooling to produce a per-band descriptor
- Two FC layers (squeeze-excite bottleneck) learn which spectral bands carry signal vs. noise
- Sigmoid output gates each band independently — noisy atmospheric bands are suppressed before any convolution

Module 2: Complementary Branch Spectral Attention (CBSA)

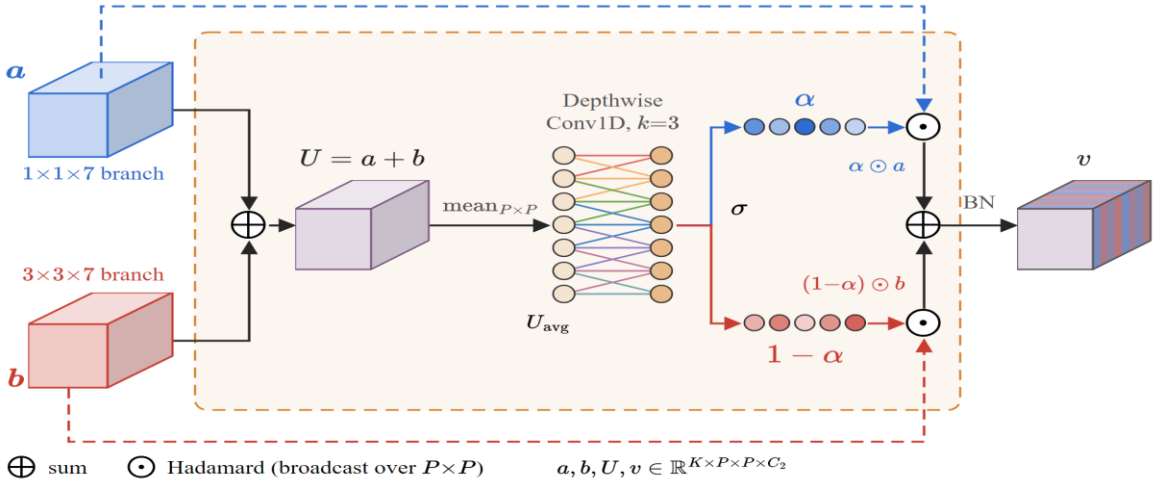
Dual-Branch Design

- Branch a: 1×1×7 Conv3D → pure spectral inter-band transitions
- Branch b: 3×3×7 Conv3D → joint spatial-spectral context
- Both: K=16 filters, spectral stride=2, BN + ReLU

Complementary Gating:

$\alpha = \sigma(\text{Conv1D}_k(\text{MeanPool}(a + b)))$ [k=3]

$v = \text{BN}(\alpha \odot a + (1-\alpha) \odot b)$



Key Insight & Efficiency

- α + (1-α) ≡ 1 → ZERO redundancy between branches
- When α → 1: spectral fidelity dominates
- When α → 0: spatial context dominates
- Forces non-overlapping orthogonal representations
- Only 80 parameters (0.03%) for the gating layer
- Followed by 2 Spectral ResBlocks + ECA-3D (7,364 params)

Diagram: CBSA Flow

- Two parallel 3D convolutions extract narrow-band (1×1×7) and wide-band (3×3×7) features simultaneously
- A learned gate α ∈ (0,1) decides how much of each branch to keep — α and (1−α) sum to 1, eliminating redundancy
- The fused output captures both pure spectral transitions and joint spatial-spectral patterns in one pass

Module 3: Second-Order Covariance Fusion Gate (SCFG)

Covariance-Based Channel Calibration

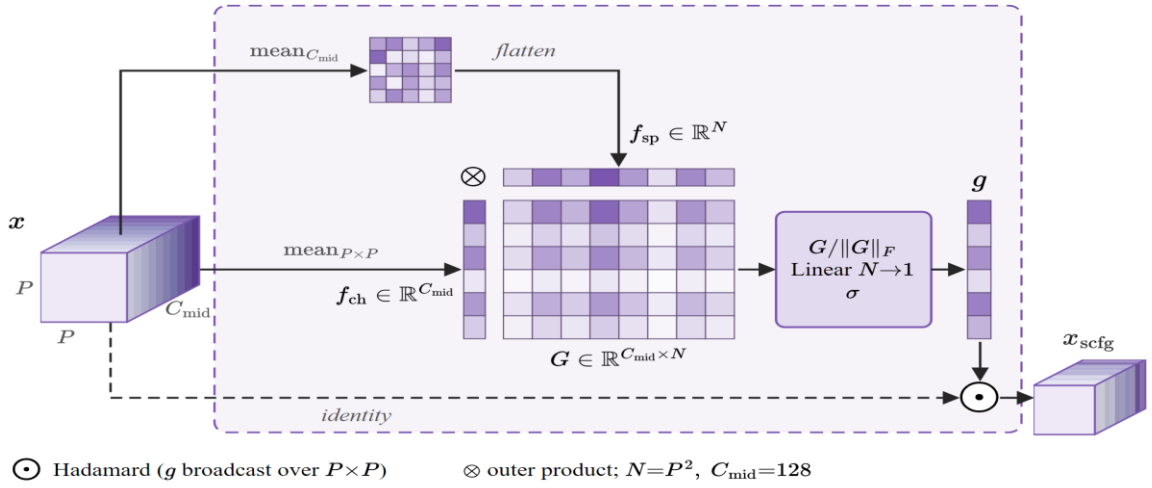
- Transitions from spectral → spatial processing stage
- Adaptive 3D pool collapses spectral dim: $C_2 \rightarrow 1$
- 1×1×1 Conv3D expands: K=16 → $C_{mid} = 128$

Second-Order Statistics:

$f_{ch} \in \mathbb{R}^{B\times C}$: channel descriptor (mean over spatial)

$f_{sp} \in \mathbb{R}^{B\times N}$: spatial descriptor (mean over channels)

$\boldsymbol{G} = f_{ch} \otimes f_{sp} \in \mathbb{R}^{B\times C\times N}$ [outer product]



Frobenius Normalization & Gating

- $\tilde{\boldsymbol{G}} = G / (\|G\|_F + \varepsilon)$ [Frobenius norm]
- $g = \sigma(W_G \cdot \tilde{G}) \in \mathbb{R}^{B\times Cmid}$ [linear projection]
- $x_{scfg} = x \odot g$ [spatial modulation]
- Only 82 parameters (0.03% of network)
- Captures bilinear correlations missed by first-order GAP

Diagram: SCFG Flow

- Channel and spatial descriptors are computed separately via global pooling, then combined through an outer product
- The resulting covariance matrix G captures second-order cross-correlations that simple averaging misses entirely
- Frobenius normalization stabilizes gradients; a linear projection produces a per-channel gate for spatial features

Minority-Aware Hybrid Loss Function

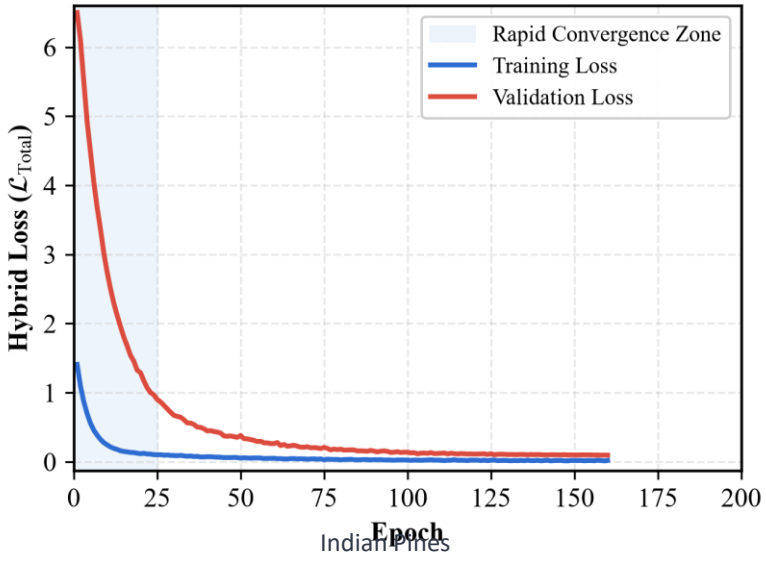
Spread Loss (weight = 0.7)

- Dynamic margin m(r) = 0.2 + 0.7·r, scheduled over training
- Early epochs: relaxed margin (m=0.2) prevents over-penalization
- Late epochs: strict margin (m=0.9) enforces wide class boundaries

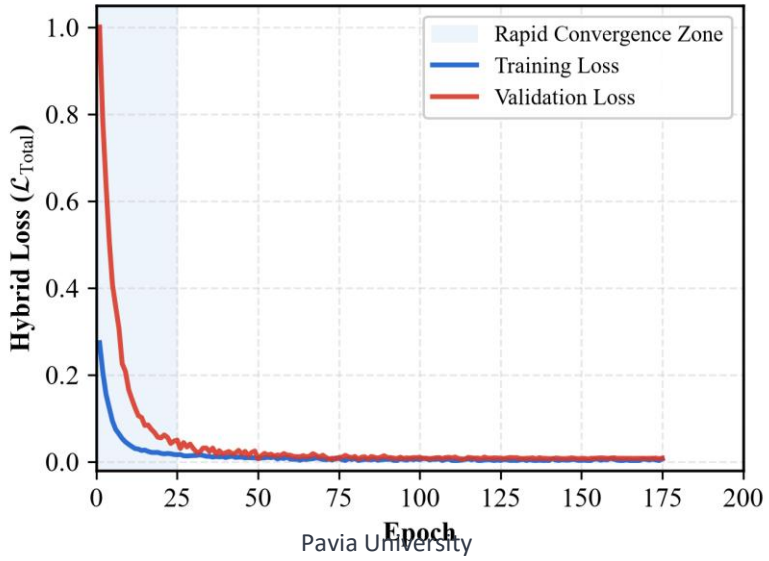
Focal Loss (weight = 0.3)

- Inverse-frequency class weights: $w_c = 1/(N_c + \varepsilon)$
- Focusing parameter γ=2.0 down-weights easy-to-classify samples
- Prevents majority classes from dominating gradients over minorities

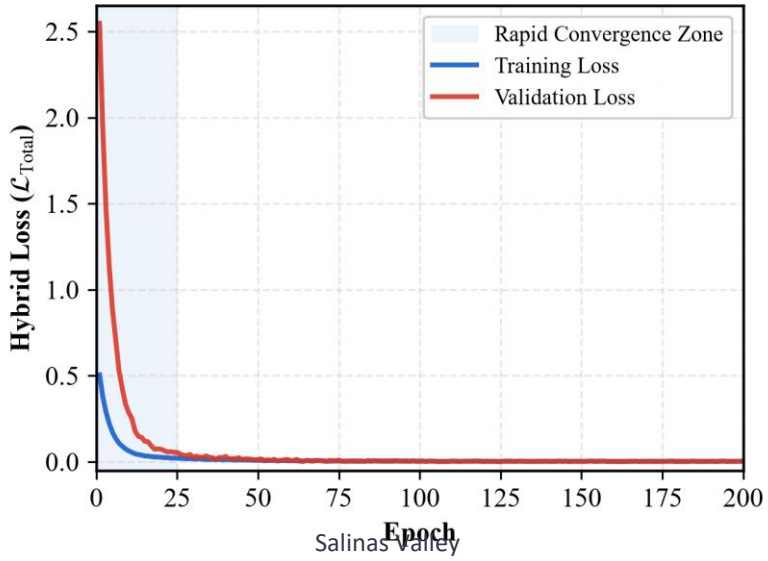
Training & Validation Loss Convergence (200 Epochs)



Indian Pines



Pavia University



Salinas Valley

Experimental Setup & Benchmark Datasets

	 Indian Pines (IN)	 Pavia University	 Salinas Scene
Sensor	AVIRIS	ROSIS	AVIRIS
Resolution	20 m/pixel	1.3 m/pixel	3.7 m/pixel
Size	145 × 145	610 × 340	512 × 217
Bands	200 (filtered)	103 channels	204 (filtered)
Classes	16	9	16
Labeled Samples	10,249	42,776	54,129

Standardized Evaluation Protocol

- Split: 10% Train / 10% Validation / 80% Test
- Stratified per class (preserved proportions)
- Patch size: 11×11 spatial neighborhood
- Batch size: 32, Epochs: 200
- Optimizer: Adam, LR: $5\times10^{-4} \rightarrow 10^{-6}$
- Cosine Annealing LR schedule
- Hardware: NVIDIA Tesla T4 GPU (Kaggle)

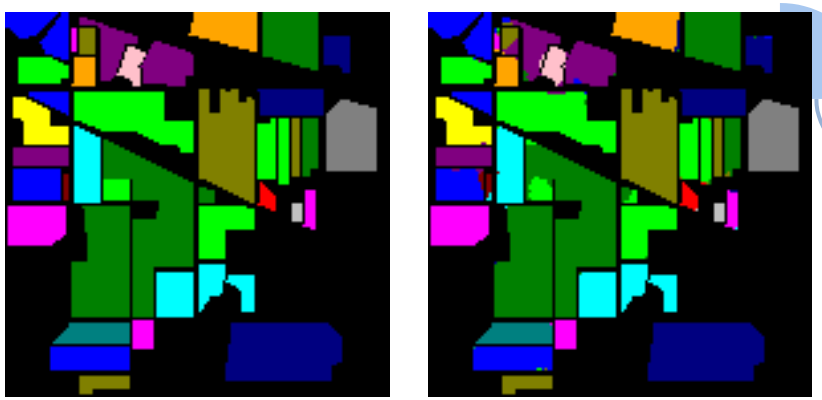
8 Baseline Models Benchmarked

- 3D CNNs: SSRN, A2S2K-ResNet
- Transformers: ViT, HSI-SSFTT, HyperRoPE-HiT
- Attention Networks: SSAN
- Capsule / Hybrid: DMC, CRV-MHDL
- All trained from scratch, identical protocol
- 24 total baseline runs (8 models × 3 datasets)

Results: Indian Pines — Per-Model Comparison

Model	OA (%)	AA (%)	κ	Time (min)
SSRN	**98.52**	79.99	0.9831	6.29
A2S2K-ResNet	98.11	94.41	0.9785	14.89
HyperRoPE-HiT	94.07	89.37	0.9324	1.43
HSI-SSFTT	98.45	94.29	0.9823	1.05
DMC	82.55	64.11	0.7989	17.73
ViT	90.33	89.12	0.8895	1.68
CRV-MHDL	67.20	51.71	0.6222	1.33
SSAN	64.19	72.77	0.5979	14.00
SGSCF-Net (Ours)	**98.05**	**94.82**	**0.9778**	**8.80**

Classification Maps



Ground Truth | SGSCF-Net

Key Finding

- SSRN tops OA (98.52%) but 0% on
- Alfalfa, Grass-mowed, Oats
- SGSCF-Net: BEST AA (94.82%)
- Alfalfa: 95.00%, Oats: 54.17%
- vs. SSRN: 0%, 0%, 0% on those

Results: Pavia University — Urban Material Classification

99.82%	99.71%	99.76	12.9 min
Overall Accuracy	Average Accuracy	Kappa (κ × 100)	Training Time



Ground Truth



SGSCF-Net

Per-Class Highlights

- 4 classes at 100.00%: Asphalt, Painted Metal, Bare Soil, Bitumen
- Meadows: 100.00% (14,919 test pixels classified perfectly)
- Gravel: 99.76% — historically confused with Asphalt
- Shadows (947 px): 99.34% via SCFG covariance operator

Baseline Context

- SSRN: 99.87% OA (marginally higher, 11.36 min)
- A2S2K-ResNet: 99.83% OA, 41.12 min (3.2× slower)
- SSAN: 76.82% OA — Gravel drops to 17.96%

Results: Salinas Valley — Fine-Grained Agriculture

99.92%	99.92%	99.91	57.7 min
Overall Accuracy	Average Accuracy	Kappa (κ × 100)	Training Time



Ground Truth



SGSCF-Net

Per-Class Highlights

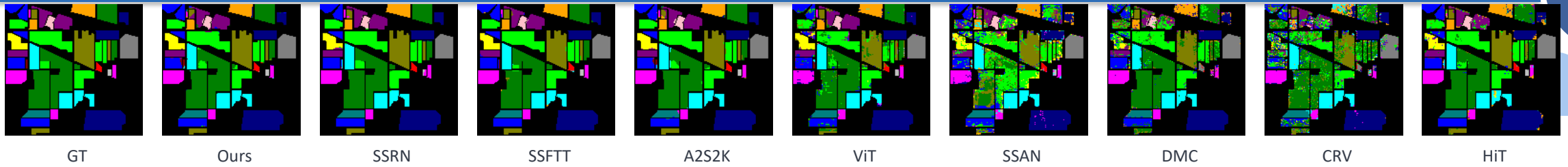
- 11 of 16 classes reach 100.00% accuracy
- All 4 Romaine lettuce growth stages (4-7 wk): 99.65-100%
- Vineyard untrained: 99.73% vs. Vineyard trellis: 100.00%
- SCFG captures chlorophyll absorption band-ratio shifts

Baseline Context

- A2S2K-ResNet: 100.00% OA (10.34 min, similar speed)
- SSAN: 88.20% OA — Grapes untrained 57.89%, collapse
- OA trend: 98.05% (IP) → 99.82% (PU) → 99.92% (SV)

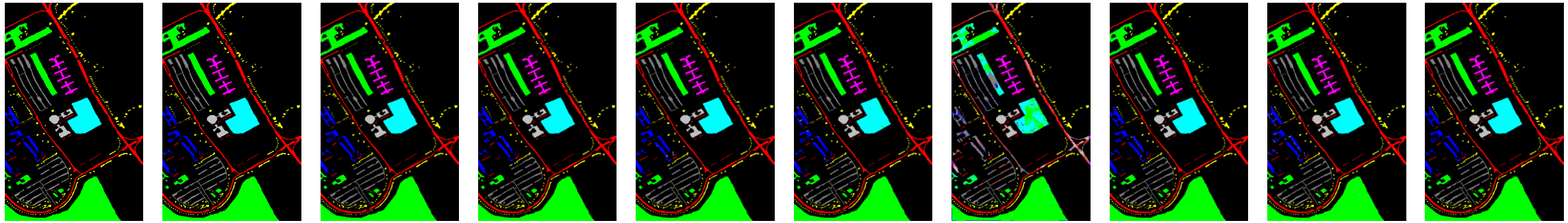
Visual Comparison: Classification Maps Across Models

Indian Pines



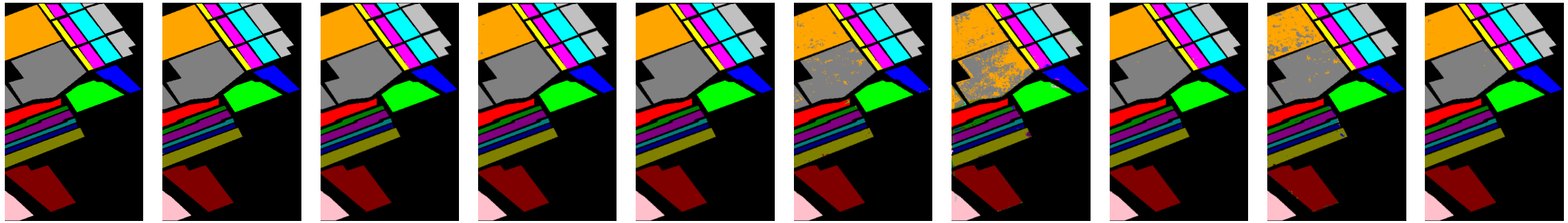
GT | Ours | SSRN | SSFTT | A2S2K | ViT | SSAN | DMC | CRV | HiT

Pavia Univ.



GT | Ours | SSRN | SSFTT | A2S2K | ViT | SSAN | DMC | CRV | HiT

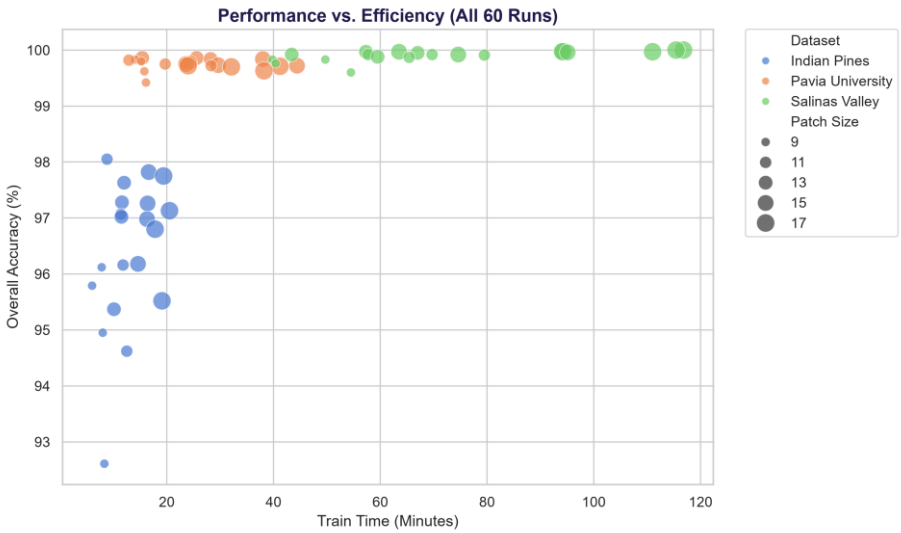
Salinas



GT | Ours | SSRN | SSFTT | A2S2K | ViT | SSAN | DMC | CRV | HiT

Computational Efficiency & Parameter Analysis

Model	Params	Checkpoint	IP Time	OA (%)
ViT	~610K	2.4 MB	1.68 min	90.33
HSI-SSFTT	~156K	625 KB	1.05 min	98.45
A2S2K-ResNet	~380K	1.5 MB	14.89 min	98.11
SSAN	6.40M	—	14.00 min	64.19
DMC	~480K	—	17.73 min	82.55
SGSCF-Net (Ours)	**308,097**	**1.2 MB**	**8.80 min**	**98.05**

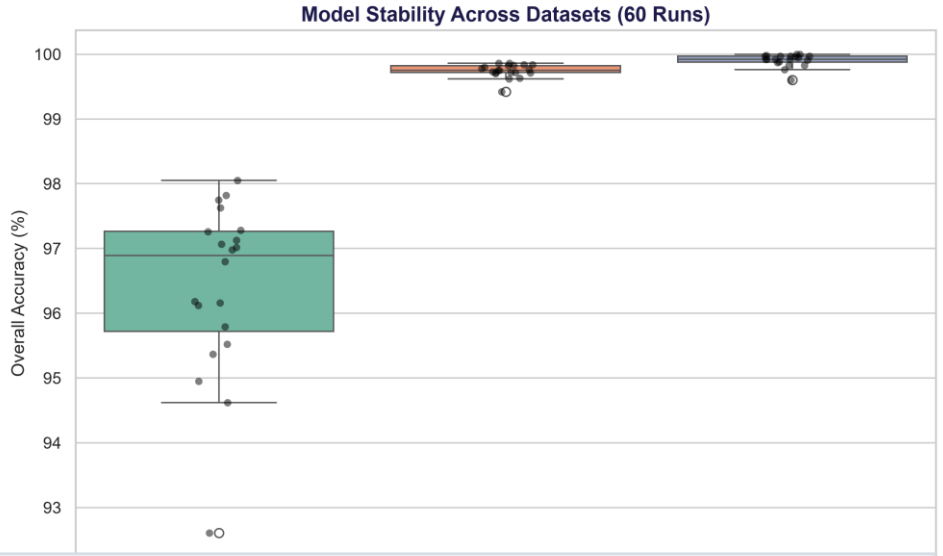


Module-Level Parameter Breakdown (Indian Pines)

- SBAM: 10,225 (3.32%) | CBSA: 80 (0.03%) | SCFG: 82 (0.03%) → 3 novel modules = only 10,387 params (3.37% of total)
- Primary volume: Spatial 3D entry (11.98%) + Spatial ResBlocks (81.00%) — standard Conv3D backbone
- Classifier head: 272 params (0.09%) — single linear layer, no deep FC stacks

Ablation Study & Block Depth Sensitivity

#	SBAM	CBSA	SCFG	H.Loss	OA (%)	AA (%)
1	✓	×	×	×	93.65	89.24
2	✓	✓	×	×	95.18	91.80
3	✓	✓	✓	×	96.44	93.05
4	✓	✓	✓	CE	97.26	93.78
5	**✓**	**✓**	**✓**	**✓**	**98.05**	**94.82**



Ablation Insights

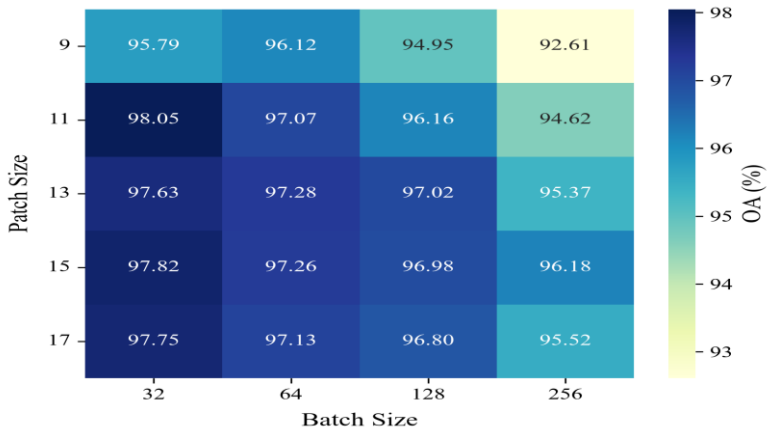
- SBAM alone lifts baseline from ~90% to 93.65% OA
- Replacing shared SE gate with CBSA: +1.53% OA
- SCFG adds +1.26% via second-order correlations
- Hybrid loss: +0.79% OA, +1.04% AA (minority protection)
- Oats class: 0% (CE-only) → 54.17% (hybrid loss)

Block Depth Sensitivity (D)

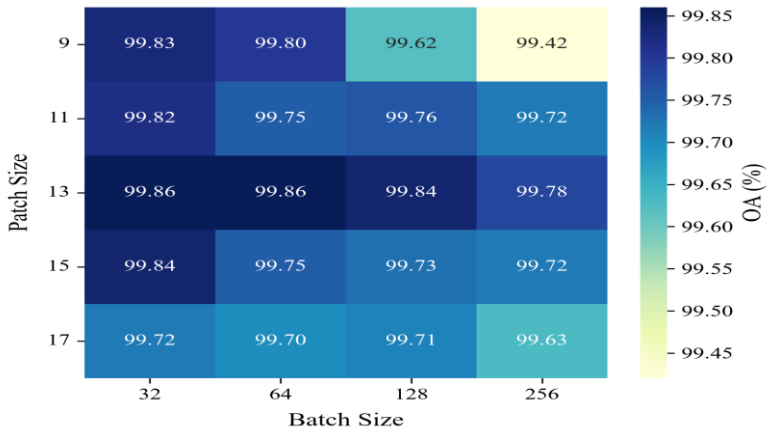
- D=1: 179K params, 95.82% OA (underfits)
- D=2: 308K params, 98.05% OA (optimal ✓)
- D=3: 437K params, 97.88% OA (diminishing returns)
- D=4: 566K params, 97.40% OA (overfitting onset)
- D=2 yields best accuracy-efficiency trade-off

60-Run Hyperparameter Grid Search

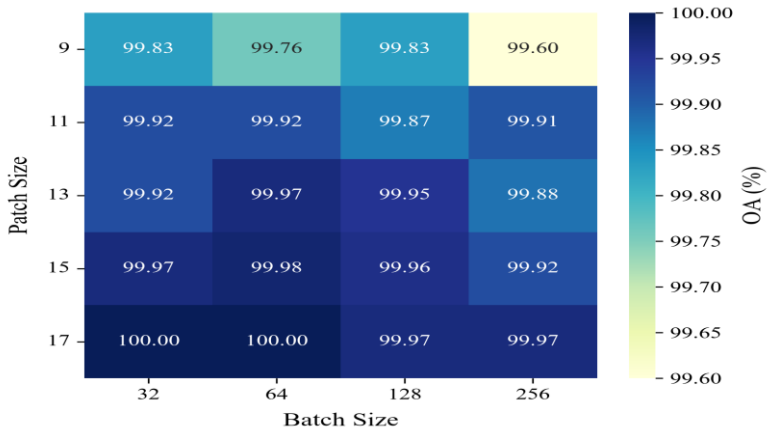
5 Patch Sizes (9×9, 11×11, 13×13, 15×15, 17×17) × 4 Batch Sizes (32, 64, 128, 256) = 20 runs/dataset × 3 datasets = 60 total



Indian Pines OA Heatmap



Pavia University OA Heatmap



Salinas Valley OA Heatmap

Optimal Configuration

- Patch 11×11, Batch 32
- OA: 98.05% (IP), 99.82% (PU), 99.92% (SV)
- Consistent Rank 1 or within 0.08%

Spatial Window Trade-off

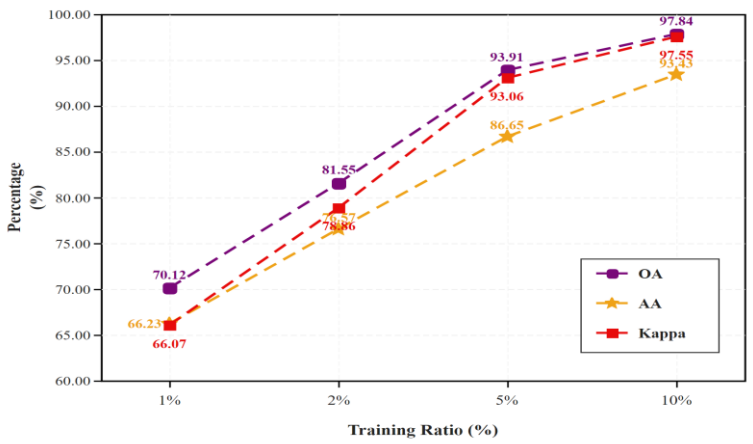
- 9×9: underfits (misses texture boundaries)
- 15×15, 17×17: overfits (cross-field contamination)
- 11×11 at 20m/pixel = 220m ground distance

Batch Size Dynamics

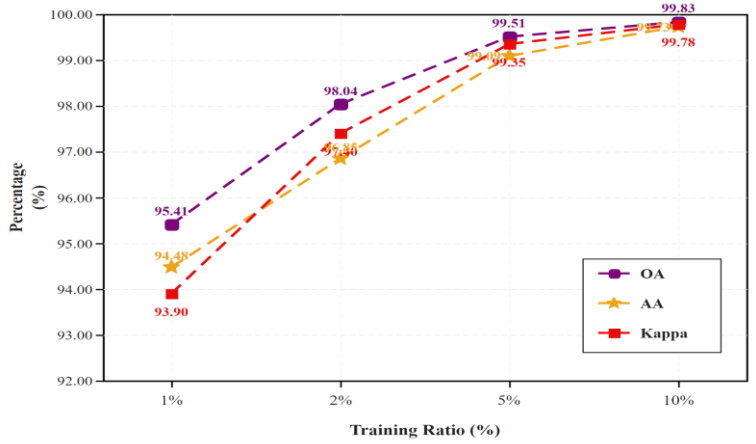
- B=32 beats B=256 by 3.43% on Indian Pines
- Small batches: higher gradient variance
- Preserves minority class decision boundaries

Data Scarcity Stress Testing: 1% – 10% Training Ratios

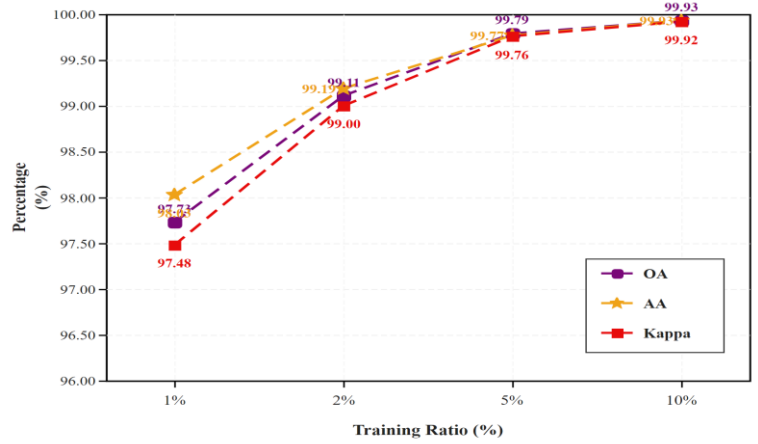
Dataset	1% Train	2% Train	5% Train	10% Train	Drop
Indian Pines	70.12%	81.55%	93.91%	97.84%	-27.73%
Pavia University	95.41%	98.04%	99.51%	99.83%	-4.43%
Salinas Valley	97.73%	99.11%	99.79%	99.93%	-2.20%



Indian Pines Degradation



Pavia University Degradation



Salinas Valley Degradation

Salinas: most resilient (−2.20%, 97.73% at 1%) | Pavia: moderate (−4.43%, 95.41% at 1%) | Indian Pines: steepest drop (−27.73%, 70.12% at 1% — Oats has 1–2 patches)

Cross-Dataset Summary & Key Achievements

98.05%	99.82%	99.92%	308K
Indian Pines OA	Pavia Univ. OA	Salinas Valley OA	Total Parameters

Project Achievements

- Designed & implemented 3 novel modules (SBAM, CBSA, SCFG)
- Best Average Accuracy on Indian Pines (94.82%) across all 9 models
- 99.82% OA on Pavia, 99.92% OA on Salinas (near-perfect)
- 120+ GPU runs: 60 grid search + 12 scarcity + 24 baseline + proposed
- O(N) linear complexity — no quadratic self-attention overhead
- 1.18s test inference on Indian Pines (vs. 135.74s SSAN)
- Complete 31-page ACM-format research manuscript drafted

Why SGSCF-Net Matters

- Lightweight: 308K params vs. 6.4M SSAN, ~610K ViT, ~480K DMC
- Balanced: Protects minority classes where SSRN/SSAN collapse to 0%
- Generalizable: Consistent top-tier across 3 diverse HSI scenes
- Efficient: 8.80 min training vs. 17.73 (DMC), 41.12 (A2S2K)
- Modular: SBAM/CBSA/SCFG are plug-and-play for any 3D CNN
- Reproducible: Fixed 10/10/80 split, 200 epochs, public datasets
- Edge-deployable: 1.2 MB checkpoint, suitable for UAV/satellite

Future Work & Research Directions

1 Hardware-Aware Quantization

Near-Term

- INT8 / FP8 tensor quantization of all Conv3D layers
- Structured channel pruning targeting <150K parameters
- Deploy on edge NPUs & FPGAs aboard UAV platforms
- Goal: real-time inference at <50ms per 11×11 patch

2 Self-Supervised Pretraining

Mid-Term

- Masked spectral autoencoder on unlabeled HSI cubes
- Fine-tune SBAM + CBSA with <1% labeled data
- Transfer across sensor modalities (AVIRIS → EnMAP)
- Goal: maintain >90% OA at 0.5% training ratio

3 Multi-Modal Sensor Fusion

Mid-Term

- Extend SCFG covariance to joint HSI + LiDAR streams
- Cross-modal attention for elevation-spectral coupling
- Benchmark on Houston 2013 & MUUFL Gulfport datasets
- Goal: exploit 3D structure for urban scene parsing

4 Journal Publication

Ongoing

- Target: IEEE Trans. Geoscience & Remote Sensing (TGRS)
- Extended experiments: statistical significance tests
- Add GAN-based augmentation ablation study
- Goal: peer-reviewed Q1 journal publication

Thank You

Questions & Discussion

SGSCF-Net: Spectral-Guided Spatial Cross-Attention Fusion Network for Hyperspectral Image Classification

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